EVERY RELATIVE CYCLE OVER A DISCRETE VALUATION RING COMES FROM A CLOSED SUBSCHEME CONSTRUCTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Surjectivity of the closed subscheme construction onto relative cycles (relative-cycles chapter, Lemma lemma-get-cycles, canonical tag 0H59). cited at proof lines 1241-1242. Primary justification: the general case already gives surjectivity.
- Equation defining the map from constructed cycles (relative-cycles chapter, Equation equation-cycle-classes, canonical tag 0H58). cited at statement line 1237. The map whose surjectivity over a DVR base the lemma asserts.
- Section on specialization of cycles (relative-cycles chapter, Section section-specialization, canonical tag 0H4H). cited at proof lines 1244-1245. Defines the closures and the construction [Zbar/X/S] used to reproduce alpha.

Source reference. Original lemma 0H59: sources/stacks-chow/tags.txt, source lines 21379–21379. Cited at proof lines 1241-1242: primary justification of the lemma.

Source reference. Original equation 0H58: sources/stacks-chow/tags.txt, source lines 21378–21378. Cited at statement line 1237: the map whose surjectivity is asserted.

Source reference. Original section 0H4H: sources/stacks-chow/tags.txt, source lines 21352–21352. Cited at proof lines 1244-1245: defines the closures and the construction [Zbar/X/S].

Lemma 0.2. *Let $f : X \to S$ be a finite type morphism of schemes with S the spectrum of a discrete valuation ring. Let $r \geq 0$. Then (0H58) is surjective.*

Proof. This of course follows from Lemma 0H59 but we can also see it directly as follows. Say α is a relative r-cycle on X/S. Write $\alpha_\eta = \sum n_i[Z_i]$ (the sum is finite). Denote $\overline{Z}_i \subset X$ the closure of Z_i as in Section 0H4H. Then $\alpha = \sum n_i[\overline{Z}_i/X/S]$. □

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.